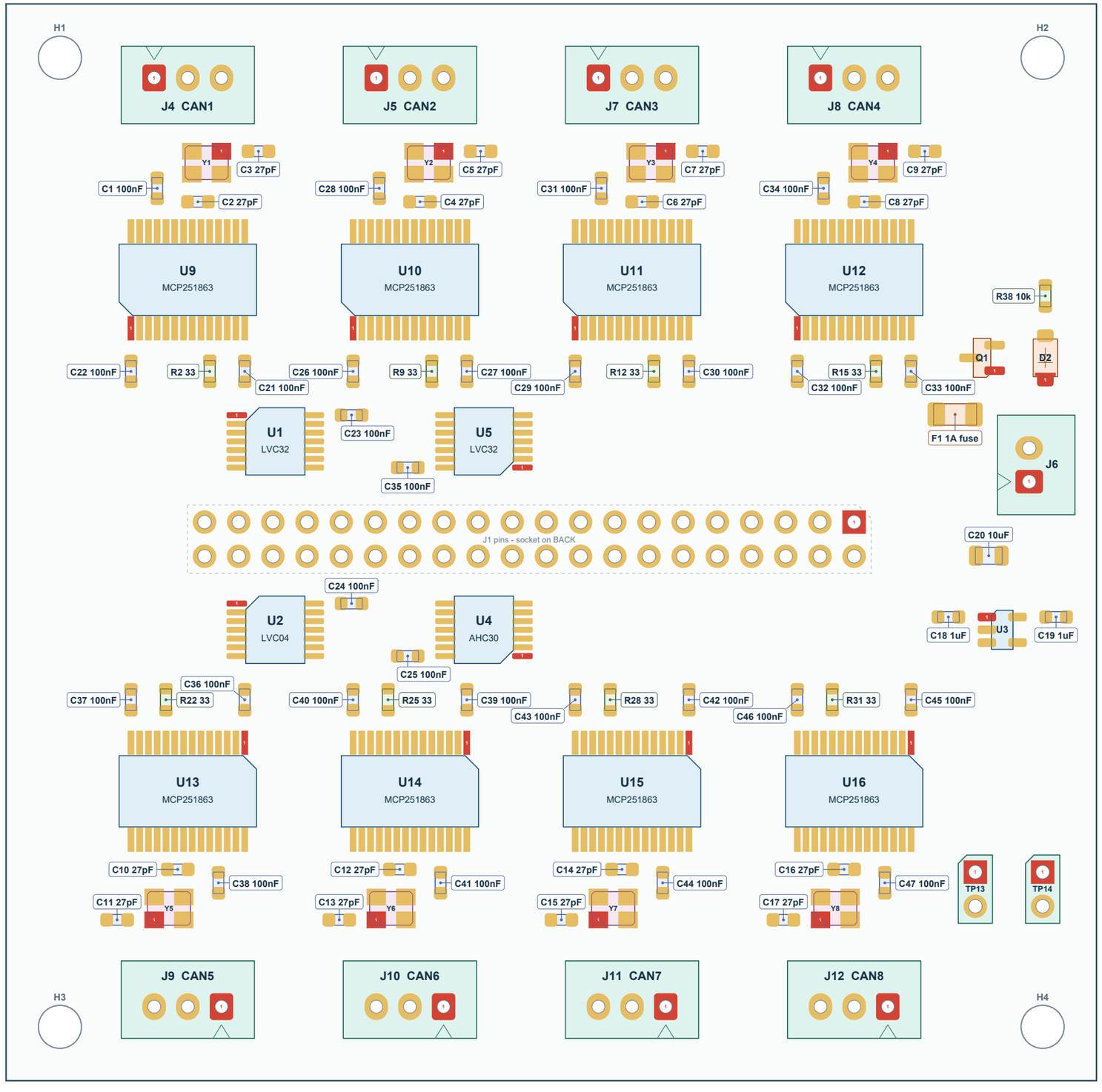
FRONT - look directly at the component face

91 components mounted on this face. Leaders point to exact footprints. Match each reference and value to the parts key.



Board top stays at the top on both views. Back view is mirrored horizontally. Zoom or print on A3; the drawing is enlarged, not 1:1.

FIT THESE PARTS ON THIS FACE

- **CAN IC | SSOP-28 x8**
  MCP251863T-E/SS
  U9, U10, U11, U12, U13, U14, U15, U16
- **40 MHz crystal | 3.2 x 2.5 mm x8**
  ABM8G-40.000MHZ-18-D2Y-T
  Y1, Y2, Y3, Y4, Y5, Y6, Y7, Y8
- **3.3 V LDO | SOT-23-5 x1**
  TLV75533PDBVR
  U3
- **27 pF | 0603 x16**
  CL10C270JB8NNNC
  C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17
- **100 nF | 0603 x28**
  CC0603KRX7R9BB104
  C1, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47
- **1 uF | 0603 x2**
  CGA3E1X7R1E105K080AC
  C18, C19
- **10 uF | 0805 x1**
  CL21B106KOQNNNE
  C20
- **10 kohm | 0603 x1**
  AC0603FR-0710KL
  R38
- **33 ohm | 0603 x8**
  RC0603FR-0733RL
  R2, R9, R12, R15, R22, R25, R28, R31
- **CAN connector | 3-pin XH x8**
  B3B-XH-A
  J4, J5, J7, J8, J9, J10, J11, J12
- **5 V input | 2-pin XH x1**
  B2B-XH-A
  J6
- **1 A fuse | 1206 x1**
  0466001.NR
  F1
- **P-channel MOSFET | SOT-23 x1**
  AO3401A
  Q1
- **6.8 V Zener | SOD-123 x1**
  BZT52C6V8-13-F
  D2
- **NAND | TSSOP-14 x1**
  74AHC30PW,118
  U4
- **OR | TSSOP-14 x2**
  74LVC32APW,118
  U1, U5
- **Inverter | TSSOP-14 x1**
  74LVC04AT14-13
  U2
- **Test header | 1 x 2 x2**
  M20-9990246
  TP13, TP14

READ THE PADS

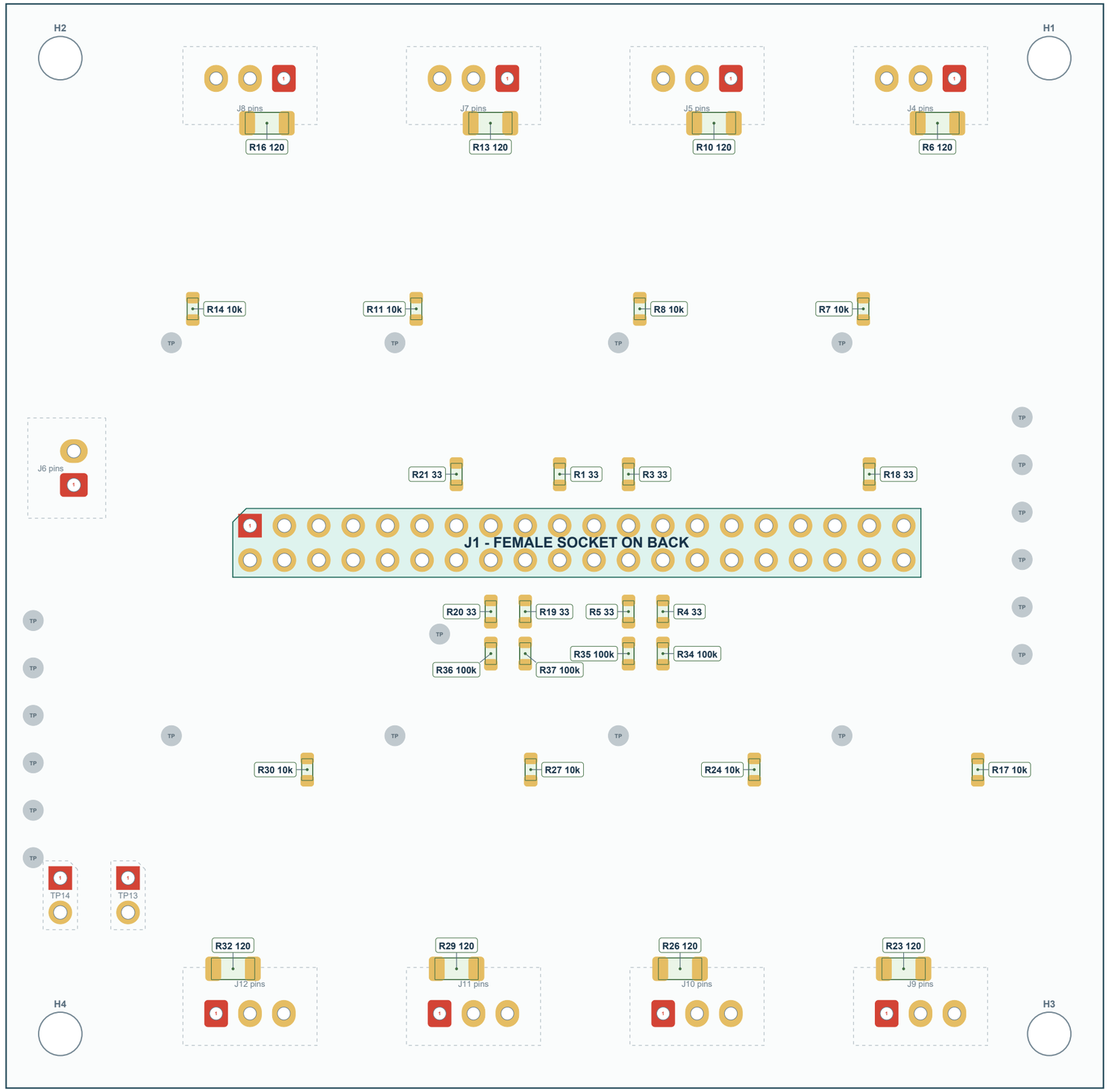
Red pad = pin 1. D2 red end = cathode band. R, C and F1 have no polarity.

Dashed J1 outline: pins only. The FEMALE socket body mounts on the BACK, and its pins are soldered from this front face.

Gray TP pads need no component. H1-H4 are mounting holes. Traces and vias are omitted for assembly clarity. Package outlines show the body position, not a photograph.

BACK - board flipped left to right

29 components mounted on this face. Leaders point to exact footprints. Match each reference and value to the parts key.



Board top stays at the top on both views. Back view is mirrored horizontally. Zoom or print on A3; the drawing is enlarged, not 1:1.

FIT THESE PARTS ON THIS FACE

- [ ] **120 ohm | 1206 x8**
  RC1206FR-07120RL
  R6, R10, R13, R16, R23, R26, R29, R32
- [ ] **10 kohm | 0603 x8**
  AC0603FR-0710KL
  R7, R8, R11, R14, R17, R24, R27, R30
- [ ] **100 kohm | 0603 x4**
  CRCW0603100KFKEA
  R34, R35, R36, R37
- [ ] **33 ohm | 0603 x8**
  RC0603FR-0733RL
  R1, R3, R4, R5, R18, R19, R20, R21
- [ ] **Jetson socket | FEMALE 2 x 20 x1**
  M20-7832046
  J1

J1 IS THE BACK-MOUNTED SOCKET

Fit the female M20-7832046 socket from this face. Its 40 legs protrude onto the front, where you solder them. Do not install J1 on the front.

In this back view, pin 1 is at the LEFT end of the UPPER row. Odd pins run left to right along the upper row; even pins run along the lower row.

OTHER THROUGH-HOLES ON THIS VIEW

Dashed JST and TP13/TP14 outlines are the solder-pad side of FRONT-mounted connectors. Their plastic bodies belong on the other face.

100K AND 10K ARE DIFFERENT PARTS

R34/R35/R36/R37: 100 kohm Vishay CRCW0603100KFKEA. R7/R8/R11/R14/R17/R24/R27/R30: 10 kohm YAGEO AC0603FR-0710KL. R38 is the remaining 10 kohm resistor, on the FRONT.

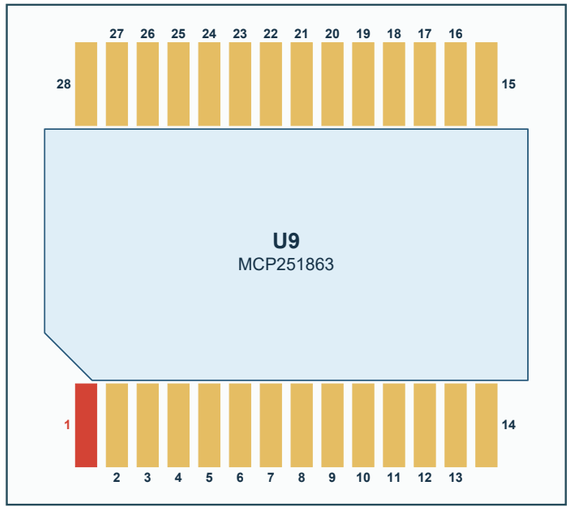
Gray TP pads need no component. H1-H4 are mounting holes. Traces and vias are omitted for assembly clarity. Package outlines show the body position, not a photograph.

Orientation checks before soldering

Every close-up retains the exact rotation of the footprint in the saved board. Red pad = pin 1.

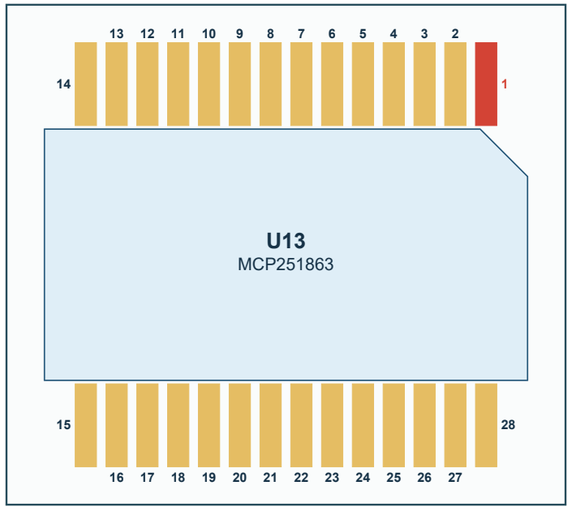
U9 - U12: upper CAN ICs

Example U9. Match the package pin-1 dot / chamfer to the red pad. Do not use the direction of the printed part name.



U13 - U16: lower CAN ICs

Example U13. These ICs are rotated 180 degrees relative to the upper group. Pin 1 moves to the opposite corner.



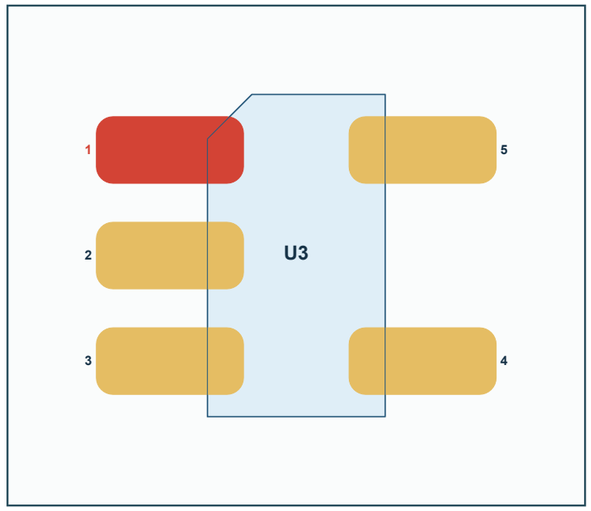
U1/U2 vs U5/U4: logic ICs

U1 and U2 share one orientation; U5 and U4 face the other way. Compare U1 (left) with U5 (right).



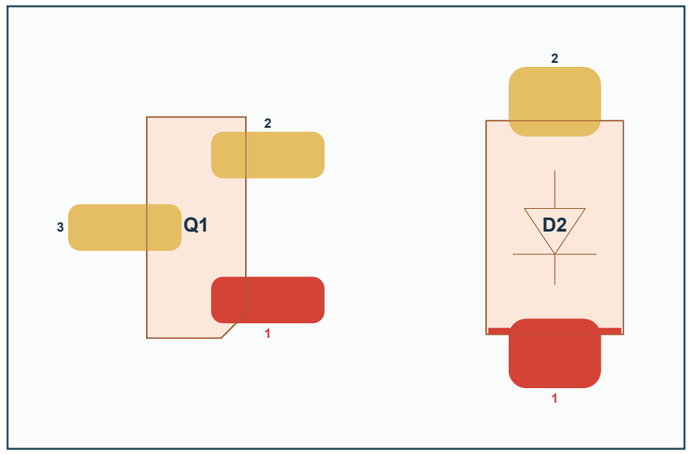
U3 - 3.3 V regulator

TLV75533PDBVR, SOT-23-5. Pin 1 = protected 5 V; pin 2 = GND; pin 3 = enable/5 V; pin 4 = NC; pin 5 = 3.3 V.



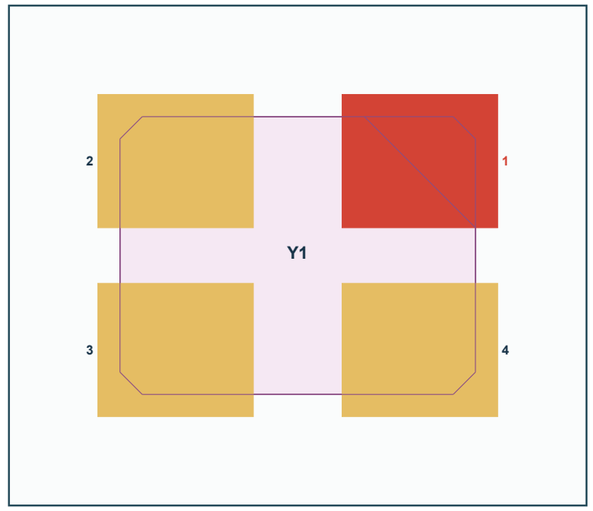
Q1 and D2 - power protection

Q1: pin 1 gate, 2 source, 3 drain. D2: pad 1 is CATHODE (K). Its band faces the lower board edge in this front view.



Y1 - Y8: four-pad crystals

Example Y1. Pins 1 and 3 are crystal terminals; 2 and 4 are GND. Keep the long axis as drawn. A 180-degree turn preserves this pattern; a 90-degree turn does not.

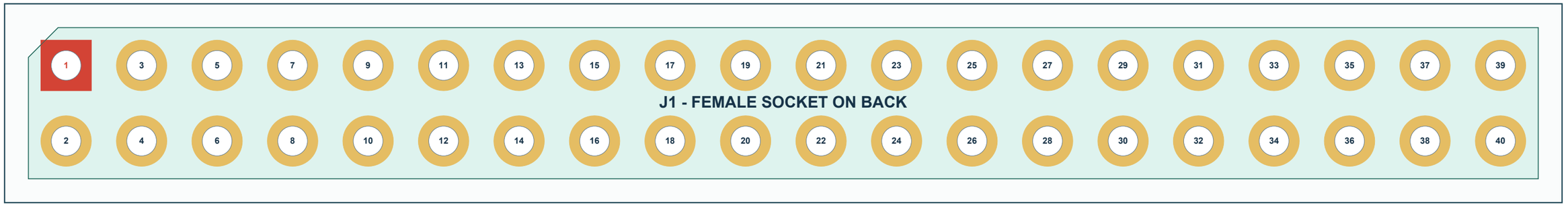


Connector placement and first voltage checks

Front and back are physical viewing directions. Red through-hole pad marks pin 1.

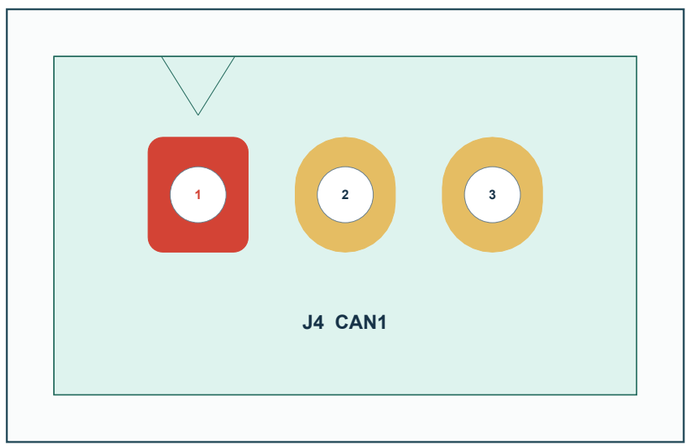
J1 - FEMALE socket mounted on the BACK

Back view, enlarged. Pin 1 is the upper-left red pad. Fit the socket from the back; solder its legs from the front. The opposite (front) view reverses left and right.



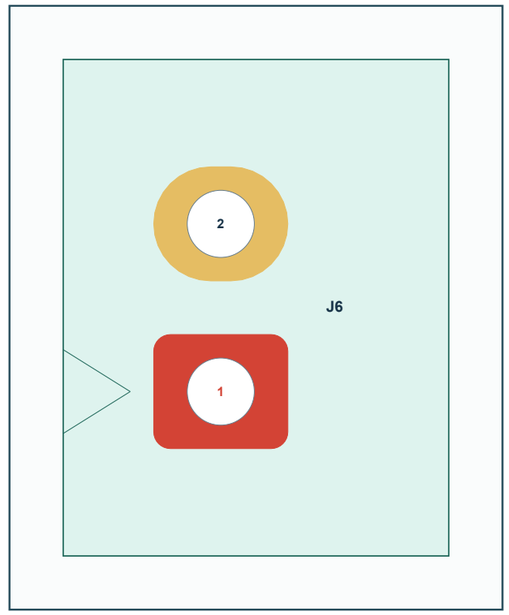
J4/J5/J7/J8/J9/J10/J11/J12

All CAN JST connectors mount on the FRONT. Example J4: pin 1 CANH; pin 2 CANL; pin 3 GND. Match the plastic housing to the outline; the connector at the other board edge is rotated.



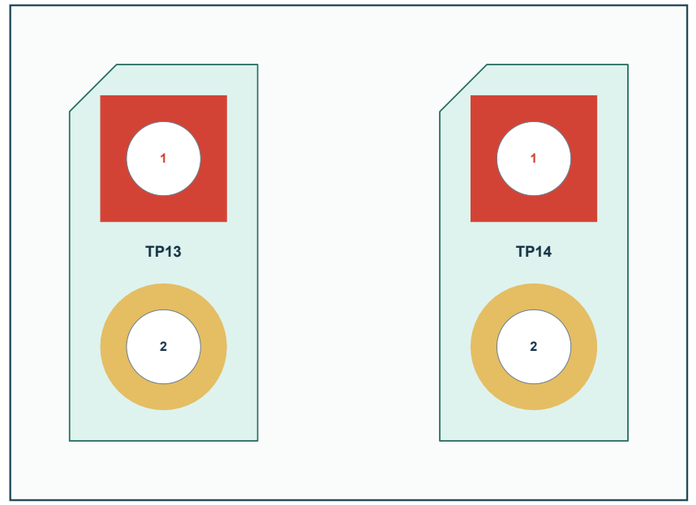
J6 - external 5 V input, FRONT

B2B-XH-A. Pin 1 = GND. Pin 2 = +5 V input BEFORE the fuse and reverse-polarity MOSFET. Confirm cable polarity against pad numbers, not wire colors.



TP13 and TP14 - test headers, FRONT

TP13: pin 1 = 3.3 V (VDD_VIO), pin 2 = GND. TP14: pin 1 = protected 5 V (VCC), pin 2 = GND. Fit M20-9990246 headers at these two locations only.



First power-up: Jetson and CAN cables disconnected. Use a current-limited 5 V supply at J6; measure TP14 then TP13 before connecting a host.

Bare TP1-TP12 and TP15-TP23 are probe pads. No header or other purchased part is fitted there.

Fitted parts key - use these exact purchased parts

120 fitted packages, 21 purchase types. Spare parts in the order list are not additional PCB placements.

Value / role	Front	Back	References	Manufacturer part to fit
CAN IC | SSOP-28	8	0	U9, U10, U11, U12, U13, U14, U15, U16	**MCP251863T-E/SS**
40 MHz crystal | 3.2 x 2.5 mm	8	0	Y1, Y2, Y3, Y4, Y5, Y6, Y7, Y8	**ABM8G-40.000MHZ-18-D2Y-T**
3.3 V LDO | SOT-23-5	1	0	U3	**TLV75533PDBVR**
27 pF | 0603	16	0	C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17	**CL10C270JB8NNNC**
100 nF | 0603	28	0	C1, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47	**CC0603KRX7R9BB104**
1 uF | 0603	2	0	C18, C19	**CGA3E1X7R1E105K080AC**
10 uF | 0805	1	0	C20	**CL21B106KOQNNNE**
120 ohm | 1206	0	8	R6, R10, R13, R16, R23, R26, R29, R32	**RC1206FR-07120RL**
10 kohm | 0603	1	8	R7, R8, R11, R14, R17, R24, R27, R30, R38	**AC0603FR-0710KL** Approved purchase substitute for RC0603FR-0710KL
100 kohm | 0603	0	4	R34, R35, R36, R37	**CRCW0603100KFKEA** Approved purchase substitute for RC0603FR-07100KL
33 ohm | 0603	8	8	R1, R2, R3, R4, R5, R9, R12, R15, R18, R19, R20, R21, R22, R25, R28, R31	**RC0603FR-0733RL**
CAN connector | 3-pin XH	8	0	J4, J5, J7, J8, J9, J10, J11, J12	**B3B-XH-A**
5 V input | 2-pin XH	1	0	J6	**B2B-XH-A**
Jetson socket | FEMALE 2 x 20	0	1	J1	**M20-7832046**
1 A fuse | 1206	1	0	F1	**0466001.NR**
P-channel MOSFET | SOT-23	1	0	Q1	**AO3401A**
6.8 V Zener | SOD-123	1	0	D2	**BZT52C6V8-13-F**
NAND | TSSOP-14	1	0	U4	**74AHC30PW,118**
OR | TSSOP-14	2	0	U1, U5	**74LVC32APW,118**
Inverter | TSSOP-14	1	0	U2	**74LVC04AT14-13**
Test header | 1 x 2	2	0	TP13, TP14	**M20-9990246**

No component is fitted on the 21 bare test pads or on H1-H4. All MLCC capacitors, resistors and F1 are non-polarized.

Keep the 27 pF and 100 nF capacitors separated on the bench; many ceramic capacitors have no readable value marking.

How to solder this board

A practical sequence for this particular double-sided, four-layer PCB.

01 Prepare parts and tools

Use magnification, fine tweezers, a temperature-controlled iron, electronics flux, fine solder wire, solder wick and fume extraction. A stencil plus paste and controlled reflow, or suitable hot-air equipment, helps with the small packages and the crystal terminals underneath the bodies. Sort each value into a labeled tray; use the fitted parts key, including the approved 10k and 100k replacements.

02 Assemble BACK SMD parts first

Fit the 28 back-side resistors. All eight 120-ohm terminations are the larger 1206 size; the 33-ohm, 10k and 100k resistors are 0603. With an iron, lightly tin one pad, hold the part flat, reflow that end, then solder the other end. Use enough heat to wet both pad and termination without prolonged contact. Leave J1 off until the small parts are finished.

03 Assemble FRONT SMD parts

Fit the 80 front-side SMD parts, checking orientation before soldering. For SSOP/TSSOP ICs, align every lead over its own pad, tack two opposite corner leads, recheck alignment, then use flux and a small solder amount to finish the pins. Remove bridges with flux and wick; inspect the heel and toe of each joint. The four-pad Y1-Y8 crystals benefit from solder paste and controlled reflow so their underside terminals actually wet.

04 Handle double-sided heating carefully

Support the PCB at its edges so already-fitted parts are not crushed. If reflowing both sides, use a process and fixture appropriate to double-sided assembly; previously soldered parts can remelt or move. Fit tall through-hole plastic connectors after the SMD reflow steps. Follow the paste and component manufacturers' profiles rather than choosing a universal hot-air or oven temperature.

05 Fit the through-hole connectors last

J4/J5/J7/J8/J9/J10/J11/J12, J6 and TP13/TP14 mount from the FRONT and are soldered from the BACK. J1 is the FEMALE 40-pin socket: fit it from the BACK and solder all 40 legs from the FRONT. Tack one pin, confirm the housing sits square and flush, then finish the remaining joints. Heat the lead and copper together and feed solder into the joint.

06 Inspect every placement before power

Check the value at every reference. Pay particular attention to 27 pF versus 100 nF, 10k versus 100k, and F1 versus a resistor. Match every IC pin-1 indicator to the red pad on the map. D2's cathode band points toward its pad 1. Q1 uses gate 1 / source 2 / drain 3. Check that no lead sits between pads, no resistor or capacitor is standing on one end, and no pin-to-pin solder bridge remains.

07 Clean and check with a meter

Remove flux residues using the flux manufacturer's recommended method. With all power and cables removed, check for unintended shorts from each supply rail to GND and between neighboring fine-pitch pins. Capacitors can briefly charge during resistance checks; a short beep alone is not a complete diagnosis. F1 is supposed to show continuity. Do not intentionally fill vias, bare test pads or mounting holes with solder.

08 Bring up the power rails first

Keep the Jetson and all CAN cables disconnected. Feed a current-limited regulated 5 V supply into J6 pin 2, with GND on J6 pin 1. Check protected 5 V at TP14 pin 1 relative to pin 2, then approximately 3.3 V at TP13 pin 1 relative to pin 2. If the supply current-limits, a part heats unexpectedly or a rail is wrong, remove power and investigate before connecting the host. The existing design still needs system timing, power-sequence and thermal validation.

Sources and drawing basis

MCP251863 pinout / package marking

Abracon ABM8G four-pad crystal

AO3401A pin assignment

TI TLV755 regulator pinout

Kester soldering and flux process guidance

Geometry and reference positions were extracted directly from the saved KiCad PCB. This guide uses the current fitted BOM, including AC0603FR-0710KL and CRCW0603100KFKEA. Neither CAD nor fabrication files were changed.

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